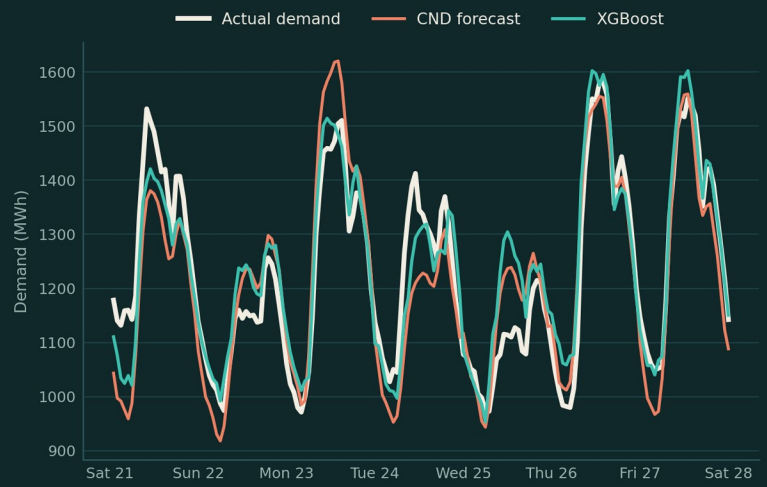
Beating the grid operator's own forecast

Short-term electricity load forecasting for the Panama national grid

Yuvraj (105001943) data and modelling lead
Chamith (105732366) documentation and project lead



Across 14 test weeks XGBoost has 22.6% lower error than CND. Week shown: 21 Dec 2019, CND 5.4% error, XGBoost 4.1%. Method: Aguilar Madrid and Antonio (2021).

Panama plans its grid a week ahead, and CND's forecast is off by about 5%

4.95%

CND's average error on the 14 test weeks [1]

9.06%

its worst week: May 2020, during the COVID lockdown [1]

Forecast too low

Risk of shortage

- Insufficient power supply
- Possible shortfalls
- Grid instability

Forecast too high

Risk of waste

- Wasted resources
- Higher operating costs
- Backup plants run for nothing

Why accuracy pays: for forecasts with 3–5% error, each 1-point cut in MAPE lowers variable generation cost by about 0.1–0.3% [2]. Hydro supplied 67% of Panama's electricity in 2020 and 42% in 2023, so planners balance rain-fed hydro against thermal plants [3, 4].

Sources: [1] Aguilar Madrid and Antonio 2021 [2] Hobbs et al. 1999 [3] Global Energy Monitor [4] IRENA

A published study already showed XGBoost can beat CND, so we reproduced it and extended it

What the research says

3.74% against 4.95%

XGBoost was the best of five models on the same 14 weeks, and beat CND in all of them [1].

One-hour-ahead is easier

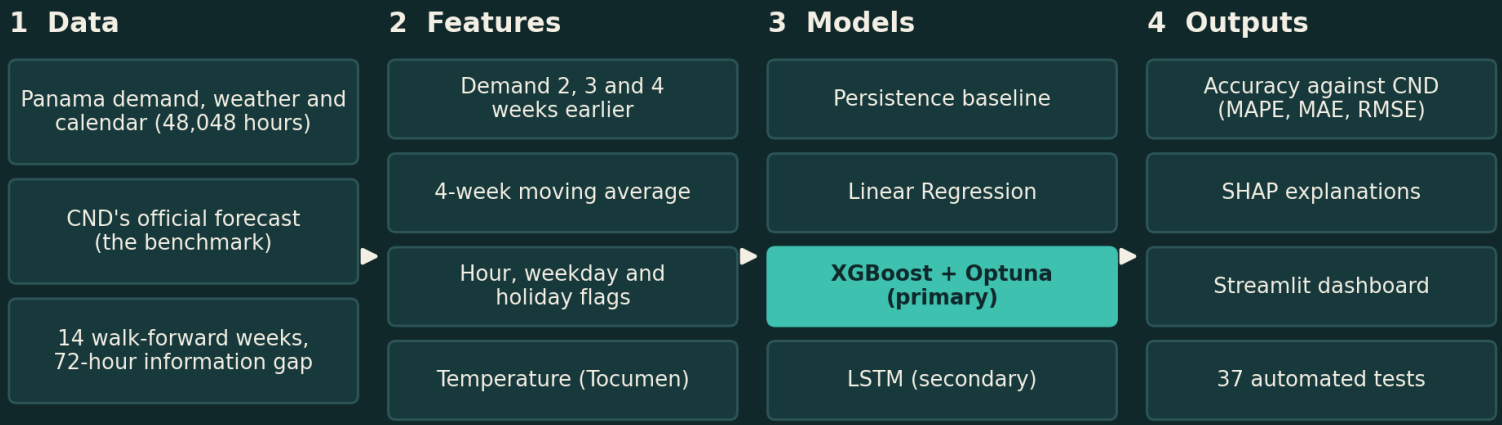
On this dataset an LSTM reaches 2.88% and XGBoost 3.26% for next-hour forecasts [5]. Ours is 168 hours ahead, so the two are not comparable.

What we add

1. Reproduce the published XGBoost settings and check them against the paper
2. Split the error by holiday, ordinary and COVID hours
3. Explain forecasts with SHAP and compare an LSTM on the same weeks
4. Deliver a working dashboard and 40 automated tests

Sources: [1] Aguilar Madrid and Antonio 2021 [5] Masfer and Dardouri 2026

XGBoost forecasts, SHAP explains, a dashboard delivers



Runs weekly: rebuild features, retrain, forecast the next 168 hours, explain, compare with CND

Data: Aguilar Madrid (2021) [6], built from CND reports, Panama Ministry of Education calendars and NASA weather. Built with pandas, scikit-learn, XGBoost [7], Optuna [9], TensorFlow, SHAP [8] and Streamlit.

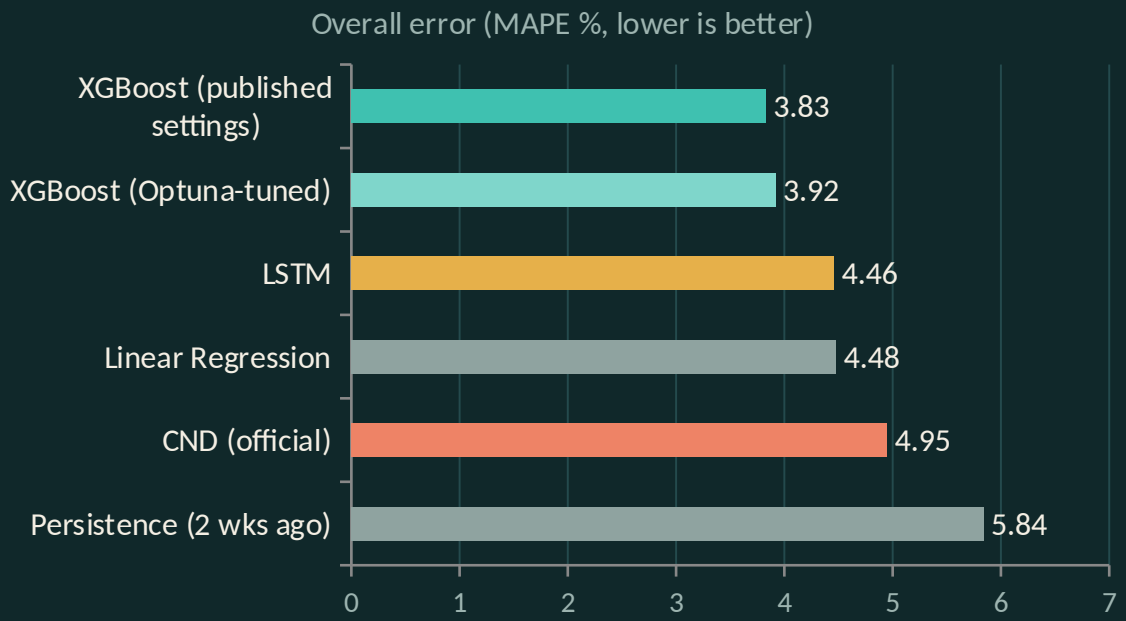
Sources: [6] Aguilar Madrid 2021 (Mendeley Data) [7] Chen and Guestrin 2016 [8] Lundberg and Lee 2017 [9] Akiba et al. 2019

Three things went against us, and we kept them in the results

1	**Fixed feature set**	We used the supplied pre-built weeks. They hold only Tocumen temperature, so other cities' weather is untested.
2	**Our tuning did not win**	Our Optuna search (20 trials) scored 3.92%. The settings published in [1] scored 3.83%, so we use those as the main model. No setting was picked after seeing the test weeks.
3	**Limited computer**	The LSTM was trained on the last 2 years with a small network so it could run on one CPU. Read its result with that in mind.

Tools: Python, pandas, scikit-learn, XGBoost, Optuna, TensorFlow, SHAP, Streamlit, pytest

XGBoost beats CND's forecast in all 14 test weeks



3.83%

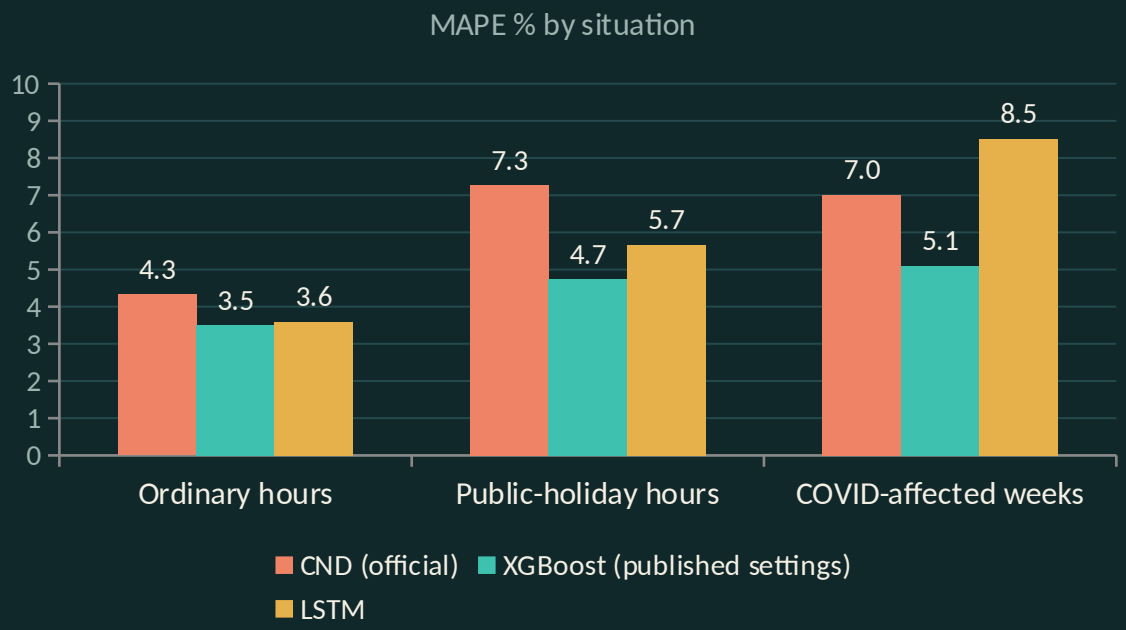
XGBoost error, against 4.95% for CND. The paper reports 3.74% [1].

46.2 MWh

average miss per hour, down from 59.2

MAPE = average percentage miss. Same 2,352 hours for every model. Source: our walk-forward evaluation; [1] for the paper's figure.

The gap is widest on public holidays and in the COVID weeks



Holidays: 4.74% vs 7.27%

35% lower, over 193 hours.

COVID: 5.10% vs 7.01%

27% lower, over 336 hours in May and June 2020.

Energy: 1.90% vs 2.81%

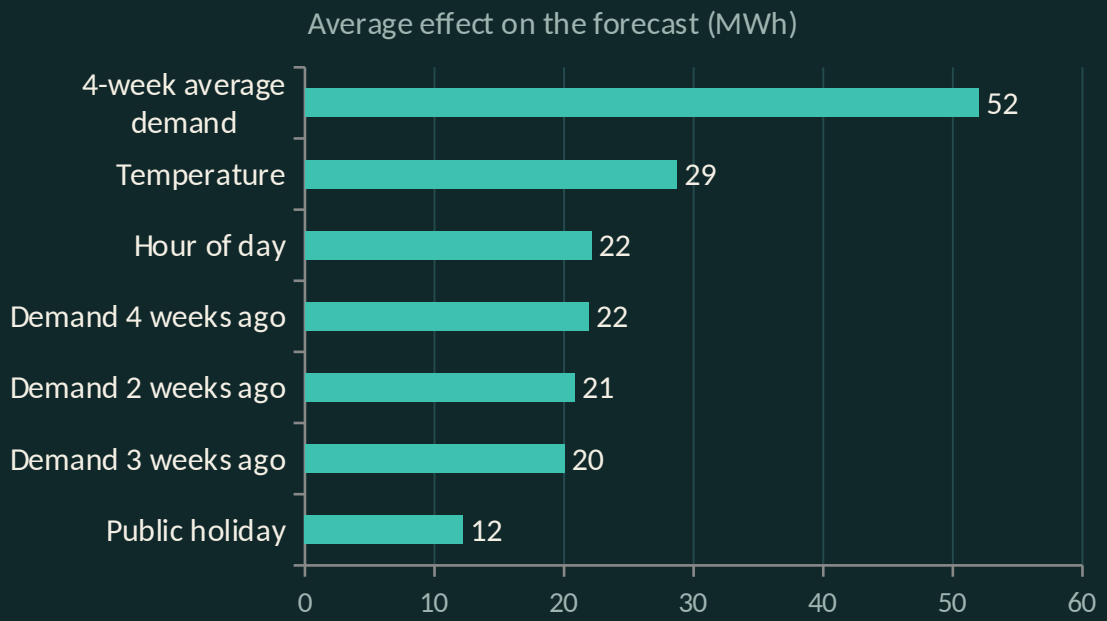
Weekly energy error, the paper's secondary metric: 32% lower [1].

LSTM, COVID: 8.5%

Worse than CND, in our compact LSTM.

Ordinary hours 1,823; holiday hours 193; COVID weeks 336. Source: our evaluation; [1] for the secondary metric.

Demand history sets the baseline; temperature, hour and holidays move it



1 **Demand history carries the most**

Four lag features add up to about 115 MWh.

2 **Temperature and hour come next**

Hotter hours push the forecast up.

3 **The holiday flag pulls holidays down**

By about 85 MWh on average.

Same ranking as the paper: demand lags first, then hour of day and temperature [1]. SHAP shows how the model uses inputs, not what causes demand.

Our numbers match the published paper, and 40 tests guard the code

Product quality: checked against the source paper [1]

Check	Paper [1]	Ours
CND error, all 14 weeks	4.95%	**4.95%**
XGBoost, published settings	3.74%	**3.83%**
Weeks XGBoost beats CND	14 of 14	**14 of 14**

CND's weekly errors match the paper's Table 5 in all 14 weeks, to two decimals. Our XGBoost is within 0.09 points of the published result.

Process quality

1. Walk-forward weeks, train on earlier data only
2. 72-hour gap, as CND works
3. 168-hour horizon (7 days)
4. Same 2,352 hours for every model
5. Holiday and COVID hours reported apart

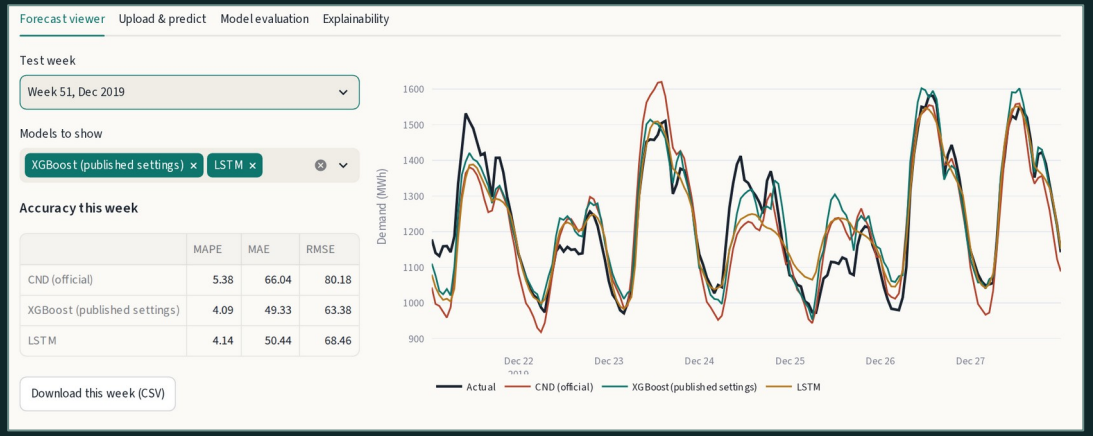
40 of 40 tests pass

16 data integrity

15 leakage checks

9 model, metrics, source check

Live demo: the dashboard, running on our real results



Real screenshot of our dashboard: the Forecast viewer tab, week of 21 Dec 2019.

1 **Forecast viewer**

Pick a test week; compare actual, CND, XGBoost and LSTM.

2 **Upload and predict**

Drop in hourly feature data, get a forecast.

3 **Model evaluation**

Accuracy overall and by situation.

4 **Explainability**

SHAP: what drives the forecast.

```
streamlit run app.py
```

Input: a test week, or a CSV of hourly rows with the 10 model features. Output: a 168-hour forecast, accuracy tables and SHAP charts.

All the code, models and tests are in one package

File	What it does
`src/`	Loads the 14 weeks; MAPE, MAE, RMSE
`run_baselines.py`	Persistence, Linear Regression, XGBoost
`run_xgb_tuned.py`	Optuna tuning, saved models
`run_lstm.py`	Compact LSTM (the slowest step)
`run_reproduction.py`	Published XGBoost settings, paper check, SHAP
`run_analysis.py`	Tables, charts, holiday and COVID split
`app.py`	Streamlit dashboard
`tests/test_qa.py`	40 automated tests
`results/`	Predictions, SHAP values, saved models

Handover

The code zip goes on Canvas, next to these slides.

Run it

```
pip install -r requirements.txt
python -m pytest
streamlit run app.py
```

Licences

Every library is permissively licensed. The dataset's own licence still needs confirming. No personal data.

There is no separate server: the backend is the pipeline plus the saved models and results, and the dashboard reads them directly.

What we did not cover, and what we would do next

Limits

Small test only 14 weeks (2,352 hours)

Weather observed, not forecast

One city only Tocumen temperature used

LSTM compact; GRU, TCN, Transformer not tried

Deployment none at CND

Roadmap (planned)

1 **Next steps**

Test on more weeks; keep holiday and shock periods separate.

2 **Integration**

Pilot beside CND with weather forecasts and all three cities.

3 **Exploration**

Tune GRU, TCN and Transformer models properly.

Questions?

Yuvraj (105001943) Chamith (105732366)